

Multimodal Extraction Benchmarks for Scholarly Documents

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Abstract

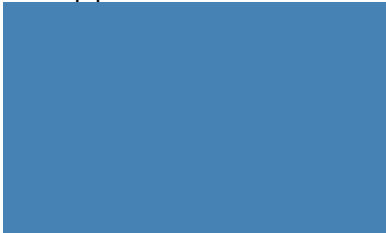
This paper reports benchmark results across three retrieval configurations, illustrated in Figure 1 and summarized in Table 1, and derives a scoring equation used throughout the evaluation.

1. Results

Table 1: Retrieval precision and recall by configuration.

Configuration	Precision	Recall
Dense only	0.71	0.65
Hybrid (dense + BM25)	0.84	0.79
Hybrid + reranker	0.91	0.88

Figure 1: Overview of the hybrid retrieval pipeline.



2. Scoring

The final relevance score combines dense and sparse signals as follows.
 $\text{score}(q, d) = a * \text{dense}(q, d) + b * \text{sparse}(q, d)$ (1)
where a and b are tunable weights fit on a held-out validation split.